Narendra Meenuga

AI/ML Engineer

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Professional Summary

AI/ML Engineer with 7 years of experience building and deploying production-grade AI/ML Platforms. I specialize in developing and deploying Enterprise-Grade domain-specific AI/ML Systems on cloud platforms and edge devices. Proficiency in building data pipelines, preparing structured inputs for ML/AI models. Expertise in ideating, implementing and deploying data-driven products by incorporating software engineering best practices, design patterns, and frameworks.

Technical Skills

Programming Languages	Python, SQL, R, Java
Frameworks & Tools	PyTorch, LangChain, LangGraph, AWS AgentCore, JAX, PaLM, CrewAI, Stable Diffusion, NeMo, NVIDIA Nim, Nemotron VLM, FastAPI, Docling , Ollama, Agentic AI.
Data Visualization	Tableau, Power BI
LLMs	Claude models (on AWS), Llama, Nemotron Parse, OpenAI models, T5, BLOOM, Bard, RoBERTa, BERT
AI/Machine Learning	Fine-tuning LLMs, Quantization, Prompt Engineering, RAG, CT-GANs, Clustering, Classification, Ensemble Models (XGboost, AdaBoost and LightGBM), Predictive Modeling, NLTK, BERT, Text Generation, Topic Modeling, Transformers
Cloud Platforms	AWS (S3, EMR, Redshift, SageMaker, Bedrock), Azure (Databricks, Synapse Analytics, AI studio), GCP (BigQuery, Vertex AI)
MLOps	MLflow, Kubeflow, Model Monitoring, Model Deployment
Databases	Oracle, MySQL, PostgreSQL, SQL Server, MongoDB, Cassandra, vector DB(pgvector & Pinecone), GraphQL
Version Control	Git, Bitbucket, Git lab
Big Data Technologies	Apache Spark, Kafka, ETL/ELT, Airflow, Databricks, Pyspark
Other Technical Skills	Docker, Kubernetes, Jenkins, Terraform & CloudFormation, shell Scripting, PowerShell

Professional Experience

Lead GenAI Engineer **CapitalOne, NY, USA**

March 2025 – Present

Responsibilities

- Developed and deployed starter kits for **RAG** and the Agentic Development Kit (**ADK**), streamlining GenAI application onboarding onto Kubernetes runtimes for enterprise tenants.
- Managed secure LLM access and Guardrails via sidecar containers, including deployment configurations, cost controls, and PCI-compliant runtime setups with S3 buckets and Kafka streams/topics for scalable batch document processing.
- Scaled a **Document Extraction** GenAI application using **Nvidia Nemotron Parse** to process millions of scanned documents for backoffice operations, delivering OCR and document intelligence capabilities tailored to customer requirements.
- Provided on-call support and runtime troubleshooting for hundreds of GenAI tenants, ensuring **application availability** and rapid issue resolution.
- Migrated manual GenAI onboarding to an automated self-service platform and shifted observability from Splunk to **Observe** by implementing an **OTEL log gateway** on GenAI clusters.
- Built **MCP-enabled CLI** tools and MCP servers that exposed internal capabilities to coding agents, accelerating self-service for library updates, workflow migrations, and diagnostics.
- Designed an **automated Docker image testing** and release framework for the converged ML platform, standardizing validation for **Kubeflow** and **Spark** workloads and reducing manual release effort.

- Developed a Python library that converts Jupyter notebooks into **Kubeflow Pipelines** and submits runs with a single command, enabling data scientists to launch experiments in hours instead of days.
- Orchestrated Jenkins-based CI/CD pipelines (with CodeGenie/LLM tooling) to build, test, scan, and promote hardened **SynapseML** Docker images, integrating unit tests, Behave suites, and security checks.
- Built optimized custom Docker images for SynapseML's **LightGBM** on Spark and contributed RFE and HPO capabilities to the internal **mlkit** library, improving model accuracy while controlling training costs.
- Implemented Kubeflow-driven **reproducibility** and **scalability** test suites that validated pipeline components across base images and cluster sizes, catching dependency issues early.
- Hardened ML infrastructure by configuring service accounts and role mappings, migrating secrets from COS to AWS Secrets Manager, and enforcing least-privilege access and privacy-aware logging.
- Troubleshot production Spark and container issues (failing jobs, performance regressions, memory bottlenecks) and applied resource tuning, caching, and image-level optimizations.

Tech Stack: Kubeflow, Docker, Kubernetes, AWS, Pyspark, LLM and GenAI tooling (Claude Code, Windsurf, MCP), Behave, Python, Vulnerability Management, Shell Scripting.

Generative AI Engineer **Symbotic Inc, MA, USA**
Feb 2024 – March 2025

Responsibilities

- Engineered robust ML risk management frameworks using **XGBoost** and scikit-learn, implementing Kubernetes-optimized distributed computing pipelines that improved model throughput by 40%.
- Established enterprise-wide model governance through comprehensive documentation protocols and CI/CD-integrated testing harnesses, reducing production incidents by 32%.
- Designed automated validation workflows leveraging Python-based statistical testing libraries, achieving 99.7% benchmark compliance across risk prediction models and optimized model performance through feature engineering and **hyperparameter tuning**, deploying containerized solutions via Kubernetes that reduced inference latency by 58%.
- Implemented risk mitigation strategies using ensemble techniques and **SHAP analysis**, enhancing model stability metrics by 27% across credit risk portfolios.
- Developed distributed data processing pipelines using Dask and Spark, enabling real-time feature generation for high-volume (1M+ TPS) fraud detection systems.
- Prototyped agentic and LLM-powered pipelines on top of internal workflow platform like DS agents that build challenger models, validation and inferencing pipelines driven by LLMs, prompt-driven workflow builders that complement existing SDK,API, UI builders, and classic-ML/Ray-based autoscaling strategies (with GPU and Polars) to scale ML components intelligently by workload.
- Built a production-scale knowledge base combining **LightRAG** and **LangGraph** for hybrid graph + vector retrieval, using Claude 3.5 Sonnet (with Haiku for cost-sensitive paths) as the reasoning backbone. Consistently hit top-percentile scores on our internal **Graph RAG** benchmarks, beating pure vector RAG by 35–48% on tricky multi-hop questions. Designed and shipped a multi-modal ingestion pipeline that actually works reliably.
- **Docling** for best-in-class parsing of PDFs, DOCX, and HTML with proper layout and table preservation, automatic knowledge graph extraction into **Neo4j**, embeddings stored in PostgreSQL(**pgvector**), and raw documents safely archived in S3.
- Performed LLM evaluations DeepEval and G-Eval.

Tech Stack: Pytorch, Scikit-learn, LangGraph, CrewAI, Autogen, DeepEval, XgBoost, ONNX, Spark, MoE, RLHF, Generative AI, vLLM, Docling, Vector Databases.

AI/ML Engineer **Synergy Technologies, IN**
Jan 2021 – Dec 2022

Responsibilities

- Boosted customer engagement by 30% with a state-of-the-art **NLP sentiment analysis** platform delivering targeted, **A/B-tested** recommendations; built using Hugging Face transformers (**BERT**, **RoBERTa**) with spaCy/PyTorch and deployed on Kubernetes via automated CI/CD pipelines.
- Shortened the model development and deployment lifecycle by 20%; implemented CI/CD automation on Azure Cloud platform to enable cross-functional teams to rapidly deploy reinforcement learning–based anomaly detection systems.
- Utilized **MLOps** tools like **MLflow**, **Kubeflow** integrated with **Azure Databricks** for model packaging, deployment, monitoring, and lifecycle management of Machine Learning models.

- Leveraged supervised, unsupervised, and reinforcement learning algorithms to develop predictive models, recommendation systems, and anomaly detection solutions, using libraries like scikit-learn, TensorFlow, and PyTorch.
- Collaborated closely with cross-functional teams to gather requirements, prototype solutions, and deploy AI systems into production, aligning with business objectives and user needs.

Tech Stack: Scikit-learn, Hugging Face Transformers, PyTorch, Azure, Databricks, SQL, Python, Pandas, NumPy, NLTK, SpaCy, Docker, Kubernetes, CI/CD Pipelines

Data Scientist/ML engineer **Cognizant Technology Solutions, IN**

Jan 2019 – Dec 2020

Responsibilities

- Boosted predictive model accuracy by 22% for customer churn prediction using XGBoost, LightGBM, and hyperparameter tuning (Bayesian optimization) while engineering high-impact features from Spark-processed datasets.
- Developed predictive models using supervised learning algorithms such as linear regression, logistic regression, decision trees, random forests, and gradient boosting machines (GBMs), achieving accuracy and performance improvements through feature engineering and model optimization.
- Employed unsupervised learning techniques, including clustering, dimensionality reduction, and anomaly detection, to identify patterns and relationships within data, uncovering hidden insights and opportunities for optimization.
- Employed cross-validation strategies such as k-fold cross-validation to assess model performance robustly, evaluating metrics such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve.
- Built scalable data pipelines and workflows using technologies like Apache Spark and Apache Airflow, orchestrating data processing tasks and ensuring reliability and efficiency in data processing and analysis.
- Designed and Implemented data processing workflows, optimizing performance, and ensuring data consistency and reliability Using Apache Spark and Apache Kafka for processing large-scale data streams and managing distributed data processing pipelines.

Tech Stack: TensorFlow, PyTorch, Kera's, Scikit-Learn, Apache Spark, Apache Kafka, AWS, SQL, Python, Pandas, NumPy

Education

Master of Science in Computer Science(AI specialization) **University of Bridgeport, Bridgeport, CT**

Bachelors in Electronics and Communication Engineering **Lovely Professional University, Punjab, IN**

Certifications

- Azure AI Engineer Associate - Certified
- AWS Certified Cloud Practitioner - Certified
- Certified Ethical Hacker (CEH) - Certified